GitHub command-line workshop guide

Use Git (git) to save versions locally and GitHub CLI (gh) to create the online repository. Open PowerShell on Windows, Terminal on macOS (zsh), or Terminal on Linux (bash). Commands apply to all three systems unless labeled otherwise.

Run one command at a time. Complete steps 1-3 before class. Use the extracted student pack and a new repository name; the supplied README.md goes into the first commit. If a command fails, resolve the error before continuing.

1. Check that both tools are installed

```
git --version
gh --version
```

Both commands should print a version number. Install any missing tools using your system below, then repeat these checks.

Windows PowerShell

```
winget install --id Git.Git -e --source winget
winget install --id GitHub.cli -e --source winget
```

Close and reopen the entire terminal application after installation. If winget is unavailable, use the Windows installers at git-scm.com/install/windows and cli.github.com.

macOS Terminal

```
brew install git gh
```

This requires Homebrew. If brew is missing, install it from brew.sh and follow its printed Next steps, then run the command above. If Git already works, brew install gh is sufficient.

Linux Terminal

Use the Linux installation appendix at the end of this guide, then return here. It includes Ubuntu/Debian and Fedora commands.

2. Sign in to GitHub

```
gh auth login --hostname github.com --git-protocol https --web
```

Follow the terminal and browser instructions to sign in to your GitHub account. If the browser does not open, open the displayed URL manually and enter the code.

3. Allow Git to use that authentication

```
gh auth setup-git
```

Confirm that the correct GitHub account is signed in:

```
gh auth status
```

GitHub CLI can now provide authentication for subsequent Git pushes.

Save the starting point locally

One partner follows steps 4-13 in the fresh student folder. All Git commands work on the three systems.

4. Move into the extracted student folder

Extract the student ZIP first. Choose the commands for your terminal below. Adjust the path if you saved the folder elsewhere or the extraction created another folder level.

Windows PowerShell

```
cd "$env:USERPROFILE\Downloads\Forensics_Exercise_Student_Pack"
Get-ChildItem
```

macOS Terminal or Linux Terminal

```
cd "$HOME/Downloads/Forensics_Exercise_Student_Pack"
ls
```

Adjust the quoted path to your extracted folder. Hidden files .gitattributes and .gitignore are included by the explicit Git commands below.

Confirm the ten project files are present: analyze_case.py, evidence_tools.py, check_solution.py, auth_log.csv, file_log.csv, evidence_manifest.json, prompts.txt, README.md, .gitattributes and .gitignore.

5. Create a local Git repository

```
git init -b main
```

This starts version tracking in the current folder and names the starting branch main.

6. Set the author details for this repository

Replace the placeholders with your name and your GitHub-associated email, or the no-reply address provided in your GitHub email settings:

```
git config user.name "Your Name"
git config user.email "YOUR_GITHUB_EMAIL"
```

These details label your commits. They apply to this repository and are separate from the login in step 2.

7. Select the ten project files for the first commit

```
git add analyze_case.py evidence_tools.py check_solution.py
git add auth_log.csv file_log.csv evidence_manifest.json
git add prompts.txt README.md .gitattributes .gitignore
```

Run each line separately. This stages the ten starter files; .gitattributes preserves the evidence CSV bytes across operating systems.

8. Check what you selected

```
git status
```

The ten files should appear under Changes to be committed. Leave teaching guides and generated output, such as audit.json, out of this first commit.

9. Record the starting point locally

```
git commit -m "Add workshop starter"
```

This saves the selected version on your computer. It has not been uploaded to GitHub yet.

Publish and verify the repository

10. Create the GitHub repository and connect it

```
gh repo create forensics-timeline-workshop --public --source=. --remote=origin
```

This public example is for the supplied fictional teaching materials. Do not add a separate README on GitHub.

If forensics-timeline-workshop already exists in your account, choose a new name such as forensics-timeline-workshop-demo in this command. The command creates the online repository and adds its connection to your local repository under the name origin.

11. Upload the first commit

```
git push -u origin main
```

This sends your saved version to GitHub. The -u option links your local main branch to the remote branch, so future uploads from this branch can use:

```
git push
```

The shorter command is for later commits; you do not need to run it again now.

12. Open the repository to check the result

```
gh repo view --web
```

Your browser opens the repository. Confirm that the ten project files are present and the README is displayed.

13. Show the saved history in the terminal

```
git log --oneline
```

You should see a short commit identifier followed by Add workshop starter. If Git opens a scrolling view, press q to return to the prompt.

Remember the three main actions

git add selects changes. git commit saves a version locally. git push uploads saved commits to GitHub.

Quick fixes

Reopen the terminal application after installing tools. For path or filename errors, revisit step 4. For createRepository token errors, check gh auth status and use browser login or a token with repository-creation access.

Official references

Installation and command references checked on 27 October 2026.

Git installation for Windows macOS and Linux

GitHub CLI installation for each operating system

GitHub Docs on adding locally hosted code

GitHub CLI manual for authentication and repositories

Linux installation appendix

Use this page only if Git or GitHub CLI is missing. Choose your distribution, finish installation, then return to the version checks in step 1. Keep the same terminal open while running each sequence. sudo may ask for your Linux password; no characters appear while you type it.

Ubuntu and Debian

These commands install Git and add the official GitHub CLI package repository. Some older distribution copies of gh have API compatibility problems.

```
sudo apt update
sudo apt install git curl
sudo mkdir -p -m 755 /etc/apt/keyrings
gh_repo="https://cli.github.com/packages"
gh_keyring="/etc/apt/keyrings/githubcli-archive-keyring.gpg"
gh_key_file=$(mktemp)
curl -fsSL "$gh_repo/githubcli-archive-keyring.gpg" -o "$gh_key_file"
sudo install -m 644 "$gh_key_file" "$gh_keyring"
```

The next command spans two lines. Paste both lines together, including the backslash at the end of the first line:

```
echo "deb [signed-by=$gh_keyring] $gh_repo stable main" | \
  sudo tee /etc/apt/sources.list.d/github-cli.list > /dev/null
sudo apt update
sudo apt install gh
```

Fedora

Use Fedora packages for both tools:

```
sudo dnf install git gh
```

For other Linux distributions, use the linked installation instructions below. Linux distributions use different package managers, so run only the commands for yours.

Git installation for Linux

GitHub CLI installation for Linux